

Assignment 5. Working with Files, Modules, and Advanced OOP

Deadline: **13.01.26**

- Each task in its own .py file
- All completed task files must be pushed to the designated GitHub repository.
- **Assessment:** Successful submission requires **GitHub push and defense**

Part 1: Working with Files and Modules

Task 1. Text File Analysis (3 points)

1. Create a program that:
 - Reads a .txt file (`text.txt`)
 - Counts:
 - Total number of lines
 - Total number of words
 - Frequency of each word
2. Save the analysis result into `analysis.txt`.

Requirements:

- Use context manager `with`
- Ignore punctuation and case sensitivity

Task 2. Working with JSON Files (3 points)

1. Create a `students.json` file containing:
 - `name`
 - `age`
 - `grades` (list)
2. Write a Python script that:
 - Reads the JSON file
 - Calculates the average grade per student
 - Writes the updated data back to a new JSON file

Requirements:

- Use `json` module

- Preserve original file

Part 2: Object-Oriented Programming

Task 3. OOP Principles & Custom Classes (3 points)

1. Create a base class `Person`
2. Create a child class `Student` that:
 - Inherits from `Person`
 - Overrides a method
3. Demonstrate:
 - Encapsulation
 - Inheritance
 - Polymorphism

Task 4. OOP Principles & Custom Classes (3 points)

1. Create a base class `Employee` with:
 - Private attribute `_salary`
 - Method `get_salary()`
 - Method `get_role()` (returns "Employee")
2. Create a child class `Manager` that:
 - Inherits from `Employee`
 - Overrides `get_role()`
 - Adds a method `get_bonus()`
3. Write a function that:
 - Accepts a list of `Employee` objects
 - Prints each employee's role and salary

Task 5. OOP Principles & Custom Classes (3 points)

- Create a class `BankAccount` with:
 - Private attributes `__balance` and `__owner`
 - Methods:
 - `deposit(amount)`
 - `withdraw(amount)`
 - `get_balance()`

- Validate input:
 - Deposit must be positive
 - Withdrawal must not exceed balance